

- 2 (a) Insulators have large band gap with a completely filled valence band and empty conduction band. At low temperature, electrons in the valence band do not have enough energy to overcome the band gap, hence, insulators cannot conduct electricity.

On the other hand, semiconductors have smaller band gap although its valence band is completely filled and conduction band empty. At higher temperature, electrons in the valence band are able to gain enough energy through thermal agitation and move into the conduction band, leaving holes behind in the valence band. Under an external applied electric field, both the electrons in the conduction band and holes in the valence band contribute to the current.

- (b) Doping is the adding of impurities into semiconductors, and as a result, the concentration of charge carriers increases which increases the conductivity. For example, in n-type doping, atoms with 5 valence electrons are added to the semiconductors. There will be more conduction electrons than holes due to the donor energy level created nearer to the conduction band. This is because the electrons from the donor level are able to move to the conduction band. On the other hand, when atoms with 3 valence electrons are added, there will be more holes due to the creation of the acceptor level nearer to the valence band, where electrons are able to transit to this level leaving holes behind in the valence band.

- 3 (a) (i) From the graph, $T = 4 \times 10 \times 10^{-3} = 0.040 \text{ s}$.